Lab 3: Component Modelling & Architectural Pattern Selection

Personal Subscription & Recurring Expense Auditor

Shreyan Ganesh SRN: PES1UG25AM813 Section: E

1. Architecture Selection and Justification

Layout: A structured subsystem-based layout is used to group related components and make their interfaces and dependencies easy to follow.

Selected Architecture: Microservices Architecture

Microservices Architecture is selected for the Personal Subscription & Recurring Expense Auditor. The system is divided into separate components for authentication, dashboard access, transaction handling, subscription management, notifications, reporting, and data storage. Each component has a specific role.

Reason 1: The system performs different types of work, such as importing transactions, analysing recurring payments, and generating reports. These services can be handled and scaled separately when required.

Reason 2: Separating the main functions makes the system easier to maintain. A problem in one service, such as the notification service, does not necessarily stop other functions such as viewing the dashboard or analysing transactions.

Security: Financial data can be kept within controlled services such as the Payment Service and Transaction Data Store. The system can use TLS for communication and encryption for stored sensitive data, while avoiding plaintext banking credentials.

Performance: Services can work independently, so transaction processing and recurring-payment analysis can be handled without unnecessarily affecting authentication or dashboard operations. This helps the system remain responsive as the amount of data increases.

2. Component Diagram

The component diagram shows the main parts of the Personal Subscription & Recurring Expense Auditor and how they communicate. The components are grouped into UserPortal, SubscriptionMgmt, and DataStore subsystems.

Components and responsibilities

Component	**Responsibility**
Authentication	Handles user authentication and user sessions.
Dashboard	Displays subscription information and reports to the user.
Payment Service	Handles imported financial transaction information.
Order Manager	Manages subscription-related operations and transaction information.
Notification Service	Handles renewal notification operations.
Spend & Cancellation Reporting	Provides spending and cancellation-related reports.
Transaction Data Store	Stores transaction information used by the system.

Main interfaces

Interface	**Purpose**
ImportTransactions	Imports transaction information into the system.
StoreTransaction	Stores imported transactions in the Transaction Data Store.
ManageSubscriptions	Connects the Dashboard with the Order Manager.
ManageTransaction Data	Allows the Order Manager to access transaction data.
ManageNotifications	Connects the Order Manager with the Notification Service.
ManageReports	Connects notification-related processing with reporting.

UML notation: The diagram uses subsystem boundaries, component notation, ports, provided interfaces (ball notation), required interfaces (socket notation), and connectors to show the relationships between components.

System flow: Transaction data is imported and stored in the Transaction Data Store. The Order Manager uses this data for subscription management. The Notification Service handles renewal notifications, while the reporting component provides spending and cancellation information. The Dashboard presents the results to the user.

3. UML Component Diagram — Visual Representation

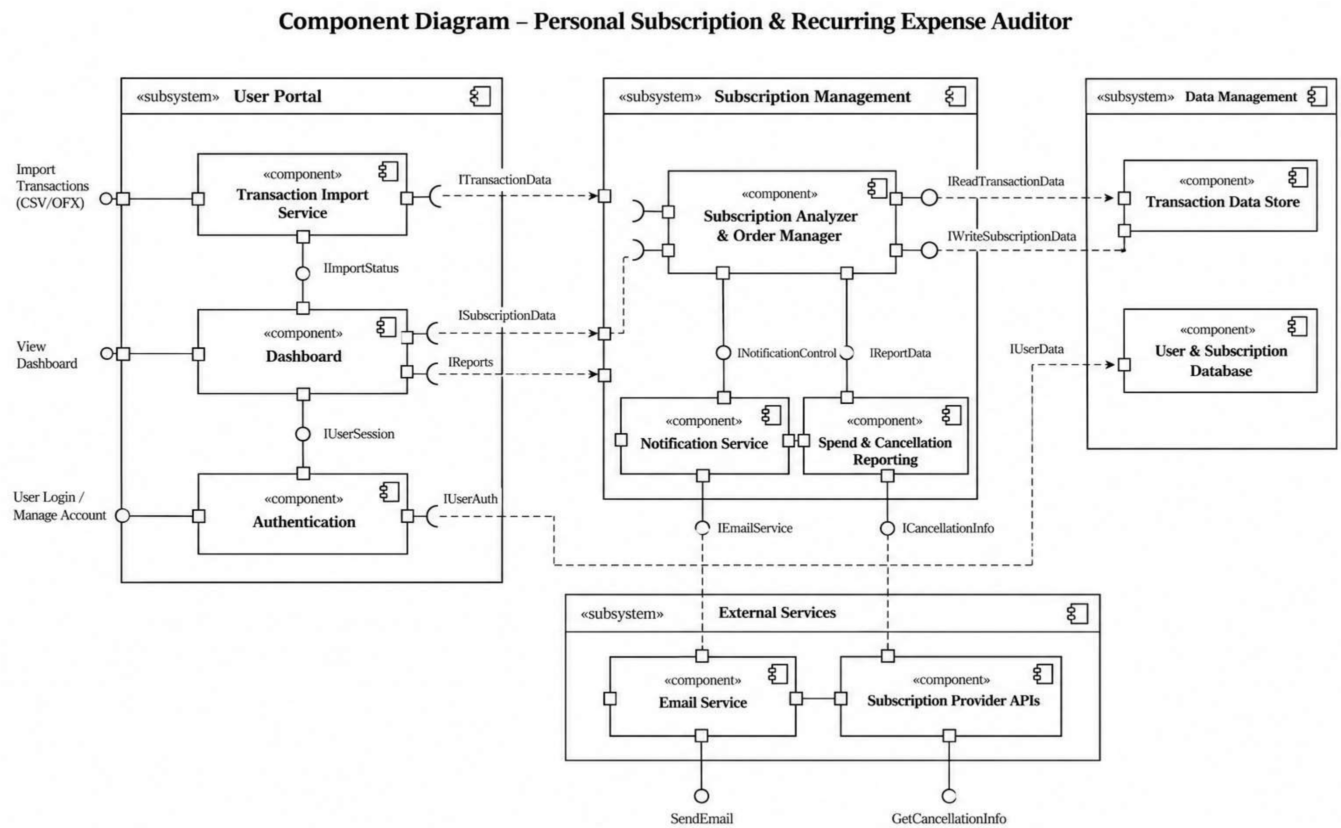

Figure 1: Component Diagram of the Personal Subscription & Recurring Expense Auditor